A Generalized Structure-Preserving Analytical Embedded Framework for Nonlinear Dynamical Systems

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***Abstract*—Physics-Informed Neural Networks (PINNs) and black-box Neural ODEs often suffer from long-term rollout instability and cumulative drift when applied to nonlinear oscillatory systems [2], [9]. Bridging the paradigms of Universal Differential Equations [8], sparse system identification [10], and mechanistic gray-box modeling, we propose a generalized Structure-Preserving Analytical Embedded framework demonstrated across the Duffing oscillator and the self-sustained Van der Pol limit-cycle oscillator [5]. By directly embedding governing analytical templates into the network architecture while treating physical coefficients as learnable tensors, our method achieves robust inverse parameter identification and stable long-term trajectory extrapolation. Empirical benchmarks confirm high-fidelity generalization, yielding robust training and test extrapolation performance across benchmark domains.**

***Index Terms*—Universal Differential Equations, Physics-Informed Neural Networks, Nonlinear Dynamics, Gray-Box Modeling, Parameter Identification.**

I. Introduction

Nonlinear dynamical systems, ranging from conservative and forced oscillators to self-sustained limit cycle generators, present major challenges for standard black-box neural networks and unconstrained Neural Ordinary Differential Equations (Neural ODEs) [6]. Unconstrained models typically accumulate rollout errors rapidly due to the violation of physical invariants and truncation error propagation.

To bridge this gap, recent advancements in Scientific Machine Learning (SciML) advocate for **Universal Differential Equations (UDEs)** [8] and gray-box modeling, where known analytical structures are hybridized with data-driven optimization. Similarly, sparse identification frameworks like **SINDy** [10] highlight the importance of preserving governing equation structures. Inspired by these foundational paradigms, we constrain the hypothesis space by embedding exact analytical templates directly into the neural architecture.

II. Generalized Methodology

Let a nonlinear dynamical system be represented by state vector $\mathbf{u} = [x_0, x_1]^T$ governed by an analytical template:

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}; \boldsymbol{\theta}) \tag{1}$$

where $\boldsymbol{\theta}$ denotes learnable physical parameters. For instance, in the Duffing oscillator, $\boldsymbol{\theta} = \{\delta, \alpha, \beta\}$ parameterized via cubic restoring forces, whereas for the Van der Pol oscillator, $\boldsymbol{\theta} = \{\mu\}$ governing limit-cycle damping:

$$\begin{aligned} \frac{dx_0}{dt} &= x_1 \\ \frac{dx_1}{dt} &= \mu(1 - x_0^2)x_1 - x_0 \end{aligned} \tag{2}$$

Optimization is performed across chronological time-series trajectories sampled via high-accuracy Runge-Kutta ($RK45$) integration schemes [7].

III. Introduction and Hybrid Methodology

To account for unmodeled dynamics, measurement noise, and structural misspecifications, purely analytical templates can be augmented with a flexible neural residual network $G_{\mathbf{w}}(\mathbf{u})$ [8], [9]. The hybrid vector field is formulated as:

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}_{\text{analytical}}(\mathbf{u}; \boldsymbol{\theta}) + G_{\mathbf{w}}(\mathbf{u}) \tag{3}$$

where $\mathbf{f}_{\text{analytical}}$ enforces primary physical constraints and learnable parameters $\boldsymbol{\theta}$, while $G_{\mathbf{w}}$ compensates for uncaptured residual errors. Regularization terms are applied to $G_{\mathbf{w}}$ to prevent overfitting and ensure structural identifiability [10].

IV. Experimental Results and Benchmarking

The generalized framework was evaluated on benchmark nonlinear systems using 70% training trajectories and 30% unseen future states for extrapolation testing.

A. Duffing Oscillator Performance

- **Metrics**: Training R^2: 0.9959 — Test Extrapolation R^2: 0.9671
- **Parameter Recovery**: True ($\delta = 0.2, \alpha = -1.0, \beta = 1.0$) vs. Learned ($\delta = 0.2175, \alpha = -1.0000, \beta = 0.9993$).

B. Van der Pol Limit Cycle Performance

- **Metrics**: Training R^2: 0.9595 (RMSE: 0.28224) — Test Extrapolation R^2: 0.8377 (RMSE: 0.52870)
- **Parameter Recovery**: True ($\mu = 1.5000$) vs. Learned ($\mu = 1.4975$).

V. Conclusion

The generalized structure-preserving analytical embedding framework—grounded in the principles of Universal Differential Equations and mechanistic modeling—successfully eliminates chaotic rollout divergence across diverse nonlinear classes. By fusing foundational physical structures with deep learning optimization, our approach provides a scalable, interpretable pathway for scientific machine learning.

References

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